Duality theory

Outline

Dual problem of an LP problem in canonical max form
- Weak Duality
- Strong Duality

Primal and dual problems: possible outcomes

Complementary Slackness
- Dual problem in matrix notation

Dual problem of a generic LP problem

Dual problem of an LP problem in canonical max form

Introduction: Dual problem of an LP problem in canonical max form

- To every LP problem, called the **primal problem**, we associate a second LP problem, called the **dual problem**
- The dual problem looks for the **best upper bound** on the optimal value of the primal problem, obtained from linear combinations of its constraints
- We define it for an LP problem in canonical maximization form and we prove that the dual of the dual is the primal problem
- Then we prove the two main results of the chapter: **Weak Duality** and **Strong Duality**

Dual problem (1/2)

Definition 1: dual problem of an LP problem in canonical maximization form

*Given a Linear Programming (LP) problem in canonical maximization form with n non-negative variables and m less-than-or-equal-to linear constraints (called the **primal problem** and denoted by LP):*

$$\max \quad \sum_{j=1}^{n} c_j \, x_j$$

$$\textit{subject to} \quad \sum_{j=1}^{n} a_{ij} \, x_j \leq b_i, \quad \forall i \in \{1, 2, \ldots, m\},$$

$$x_j \geq 0, \quad \forall j \in \{1, 2, \ldots, n\},$$

...

Dual problem (2/2)

Definition: dual problem of an LP problem in canonical maximization form

. . . the following LP problem with m non-negative variables and n greater-than-or-equal-to linear constraints

$$\min \quad \sum_{i=1}^{m} b_i \, \pi_i$$

$$\text{subject to} \quad \sum_{i=1}^{m} a_{ij} \, \pi_i \geq c_j, \quad \forall j \in \{1, 2, \ldots, n\},$$

$$\pi_i \geq 0, \quad \forall i \in \{1, 2, \ldots, m\},$$

is called the ***dual problem*** *and denoted by* $D(LP)$.

- The primal problem has n variables and m constraints; the dual problem has m variables and n constraints
- Each **constraint** i of the primal problem gives one **dual variable** π_i, and each **variable** x_j of the primal problem gives one **dual constraint**

Let us write the dual problem
of a small LP problem with two variables.

Example (1/2)

- Consider the following primal problem LP in canonical maximization form with $n = 2$ non-negative variables and $m = 3$ less-than-or-equal-to linear constraints (the **guiding example** of this chapter):

$$\begin{array}{rrcrcr} \max & 5\,x_1 & + & 2\,x_2 & & \\ \text{subject to} & -\,x_1 & + & 3\,x_2 & \leq & 12 \\ & x_1 & + & 2\,x_2 & \leq & 12 \\ & 3\,x_1 & - & 2\,x_2 & \leq & 8 \\ & x_1, & & x_2 & \geq & 0 \end{array}$$

Example (2/2)

- The dual problem $D(LP)$, with $m = 3$ non-negative variables and $n = 2$ greater-than-or-equal-to linear constraints, reads as follows:

$$\begin{array}{rrcrcrcl} \min & 12\,\pi_1 & + & 12\,\pi_2 & + & 8\,\pi_3 & & \\ \text{subject to} & -\,\pi_1 & + & \pi_2 & + & 3\,\pi_3 & \geq & 5 \\ & 3\,\pi_1 & + & 2\,\pi_2 & - & 2\,\pi_3 & \geq & 2 \\ & \pi_1, & & \pi_2, & & \pi_3 & \geq & 0 \end{array}$$

- The right-hand sides $12, 12, 8$ of the primal problem become the objective function coefficients of the dual problem, and the objective function coefficients $5, 2$ become its right-hand sides

Where does the dual problem come from?

It comes from the search for
upper bounds on the optimal value $z(LP)$.

Upper bounds from the constraints

- Upper bounds on the optimal value $z(LP)$ of the primal problem can be obtained by taking **non-negative linear combinations** of the constraints
- The multipliers are chosen so that the coefficient of each variable in the resulting left-hand side is **greater than or equal to** its coefficient in the objective function
- The right-hand side of the resulting inequality then provides an **upper bound** on $z(LP)$

Example: upper bounds from linear combinations (1/4)

- Let us consider the primal problem of the guiding example:

$$\begin{array}{rrcrcr} \max & 5\,x_1 & + & 2\,x_2 & & \\ \text{subject to} & -\,x_1 & + & 3\,x_2 & \leq & 12 \\ & x_1 & + & 2\,x_2 & \leq & 12 \\ & 3\,x_1 & - & 2\,x_2 & \leq & 8 \\ & x_1, & & x_2 & \geq & 0 \end{array}$$

- We try three different non-negative linear combinations of its constraints

Example: upper bounds from linear combinations (2/4)

- For example, by multiplying the second constraint by 5, we get:

$$5\ x_1 + 10\ x_2 \quad \leq \quad 60$$

- Hence, any feasible solution of the primal problem satisfies this inequality, which is an overestimate of the objective function, since we have:

$$5\ x_1 + 2\ x_2 \quad \leq \quad 5\ x_1 + 10\ x_2 \quad \leq \quad 60$$

- In particular, these inequalities hold for the optimal solution, so

$$z(\ LP\ ) \leq 60$$

Example: upper bounds from linear combinations (3/4)

- For example, by multiplying the second constraint by $\frac{5}{2}$ and then adding it to the third constraint, we get:

$$\frac{5}{2}(x_1 + 2\,x_2) + (3\,x_1 - 2\,x_2) \leq 12 \cdot \frac{5}{2} + 8 \quad \Rightarrow \quad \frac{11}{2}\,x_1 + 3\,x_2 \leq 38$$

- Hence, any feasible solution of the primal problem satisfies this inequality, which is an overestimate of the objective function, since we have:

$$5\,x_1 + 2\,x_2 \quad \leq \quad \frac{11}{2}\,x_1 + 3\,x_2 \quad \leq \quad 38$$

- In particular, these inequalities hold for the optimal solution, so

$$z(\,LP\,) \leq 38$$

Example: upper bounds from linear combinations (4/4)

- For example, by multiplying the first, second, and third constraints by 2, 1, and 2, respectively, and then summing them up, we get:

$$2\,(-x_1 + 3\,x_2) + (x_1 + 2\,x_2) + 2\,(3\,x_1 - 2\,x_2)$$

$$\leq 12 \cdot 2 + 12 + 8 \cdot 2 \quad \Rightarrow \quad 5\,x_1 + 4\,x_2 \leq 52$$

- Hence, any feasible solution of the primal problem satisfies this inequality, which is an overestimate of the objective function, since we have:

$$5\,x_1 + 2\,x_2 \quad \leq \quad 5\,x_1 + 4\,x_2 \quad \leq \quad 52$$

- In particular, these inequalities hold for the optimal solution, so

$$z(\,LP\,) \leq 52$$

Three choices, three different upper bounds: 60, 38 and 52.

Instead of guessing the multipliers,
let us treat them as **unknowns**.

Example: the best upper bound (1/3)

- Let us consider again the primal problem of the guiding example
- Since we have three constraints, we introduce three multipliers: π_1, π_2, and π_3
- Multiplying each constraint by its multiplier, we get:

$$(-x_1 + 3\,x_2)\,\pi_1 \leq 12\,\pi_1$$

$$(x_1 + 2\,x_2)\,\pi_2 \leq 12\,\pi_2$$

$$(3\,x_1 - 2\,x_2)\,\pi_3 \leq 8\,\pi_3$$

Example: the best upper bound (2/3)

- Summing the three inequalities, we get:

$$\begin{aligned}&(-x_1 + 3\,x_2)\,\pi_1 + (x_1 + 2\,x_2)\,\pi_2 + (3\,x_1 - 2\,x_2)\,\pi_3\\ &\leq 12\,\pi_1 + 12\,\pi_2 + 8\,\pi_3\end{aligned}$$

- Grouping by variable, we get:

$$\begin{aligned}&(-\pi_1 + \pi_2 + 3\,\pi_3)\,x_1 + (3\,\pi_1 + 2\,\pi_2 - 2\,\pi_3)\,x_2\\ &\leq 12\,\pi_1 + 12\,\pi_2 + 8\,\pi_3\end{aligned}$$

Example: the best upper bound (3/3)

- The right-hand side of this inequality provides a valid upper bound on $z(LP)$ if and only if the coefficient of each variable is at least as large as the corresponding coefficient in the objective function:

$$-\pi_1 + \pi_2 + 3\,\pi_3 \geq 5$$

$$3\,\pi_1 + 2\,\pi_2 - 2\,\pi_3 \geq 2$$

- Each multiplier must be non-negative, otherwise the direction of the corresponding inequality would be reversed:

$$\pi_1,\ \pi_2,\ \pi_3 \geq 0$$

- Since we want the best possible upper bound, i.e., the smallest one, we must **minimize** the right-hand side $12\,\pi_1 + 12\,\pi_2 + 8\,\pi_3$

The dual problem as the best upper bound

The dual problem $D(LP)$ is the LP problem
of finding the **best possible upper bound**
on the optimal value $z(\,LP\,)$ of the primal problem LP
by using linear combinations of the constraints.

- In the guiding example, the three conditions on π_1, π_2, π_3 and the objective function to minimize are exactly the dual problem $D(LP)$ written above

An important property of the duality transformation
is that it gives back the starting problem
if applied **twice**.

The dual of the dual

Observation 1

The dual problem of the dual problem is the primal problem.

- The idea of the proof: we rewrite the dual problem in canonical maximization form, we write its dual problem, and we rewrite the result in maximization form

Proof (1/4)

- Starting from the dual problem,

$$\min \quad \sum_{i=1}^{m} b_i \, \pi_i$$

$$\text{subject to} \quad \sum_{i=1}^{m} a_{ij} \, \pi_i \geq c_j, \quad \forall j \in \{1, 2, \ldots, n\},$$

$$\pi_i \geq 0, \quad \forall i \in \{1, 2, \ldots, m\},$$

- consider the equivalent LP problem obtained by multiplying the objective function and each constraint by -1 (next slide)

Proof (2/4)

- The equivalent LP problem obtained by multiplying the objective function and each constraint by -1 is:

$$\max \quad \sum_{i=1}^{m} (-b_i)\, \pi_i$$

$$\text{subject to} \quad \sum_{i=1}^{m} (-a_{ij})\, \pi_i \leq -c_j, \quad \forall j \in \{1, 2, \ldots, n\},$$

$$\pi_i \geq 0, \qquad \forall i \in \{1, 2, \ldots, m\}.$$

- This problem is now in **canonical maximization form**

Proof (3/4)

- Hence, its dual is

$$\min \quad \sum_{j=1}^{n} (-c_j)\, x_j$$

$$\text{subject to} \quad \sum_{j=1}^{n} (-a_{ij})\, x_j \geq -b_i, \quad \forall i \in \{1, 2, \ldots, m\},$$

$$x_j \geq 0, \qquad \forall j \in \{1, 2, \ldots, n\}.$$

- Its variables x_j correspond to the n constraints of the problem of the previous slide, and its m constraints to the variables π_i

Proof (4/4)

- Finally, multiplying the objective function and each constraint by -1, we obtain the equivalent problem

$$\max \quad \sum_{j=1}^{n} c_j \, x_j$$

$$\text{subject to} \quad \sum_{j=1}^{n} a_{ij} \, x_j \leq b_i, \quad \forall i \in \{1, 2, \ldots, m\},$$

$$x_j \geq 0, \quad \forall j \in \{1, 2, \ldots, n\},$$

  which is exactly the primal problem
- Therefore, the dual of the dual problem is the primal problem

□

The following theorem, called **Weak Duality**,
states that the value of any feasible solution
of the primal problem is smaller than or equal to
the value of any feasible solution of the dual problem.

Weak Duality

Theorem 1: Weak Duality

For any feasible solution $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n)$ *of the primal problem and any feasible solution* $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ *of the dual problem, we have*

$$\sum_{j=1}^{n} c_j \bar{x}_j \leq \sum_{i=1}^{m} b_i \bar{\pi}_i.$$

- The idea of the proof is to use the primal and dual feasibility of the two solutions to show that the value of the primal solution is at most equal to the value of the dual solution

Proof

We have:

$$\sum_{j=1}^{n} c_j \, \bar{x}_j \overset{(a)}{\leq} \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} \, \bar{\pi}_i \right) \bar{x}_j$$

$$= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} \, \bar{x}_j \right) \bar{\pi}_i$$

$$\overset{(b)}{\leq} \sum_{i=1}^{m} b_i \, \bar{\pi}_i$$

(a) by the dual constraints and the primal non-negativity constraints: $c_j \leq \sum_{i=1}^{m} a_{ij} \bar{\pi}_i$ and $\bar{x}_j \geq 0$, $\forall j \in \{1, 2, \ldots, n\}$

(b) by the primal constraints and the dual non-negativity constraints: $\sum_{j=1}^{n} a_{ij} \bar{x}_j \leq b_i$ and $\bar{\pi}_i \geq 0$, $\forall i \in \{1, 2, \ldots, m\}$

Weak Duality: bounds

- The value of any feasible solution $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ of the dual problem provides a valid **upper bound** on the optimal value $z(\, LP\,)$ of the primal problem:

$$z(\, LP\,) \leq \sum_{i=1}^{m} b_i \; \bar{\pi}_i$$

- The value of any feasible solution $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n)$ of the primal problem provides a valid **lower bound** on the optimal value $z(\; D(LP)\;)$ of the dual problem:

$$\sum_{j=1}^{n} c_j \; \bar{x}_j \leq z(\; D(LP)\;)$$

Let us compute some lower bounds and upper bounds for the guiding example.

Example: primal and dual feasible solutions (1/5)

- Consider the following pair of primal and dual LP problems:

$$\begin{aligned} \max \quad & 5x_1 + 2x_2 \\ \text{subject to} \quad & -x_1 + 3x_2 \leq 12 \\ & x_1 + 2x_2 \leq 12 \\ & 3x_1 - 2x_2 \leq 8 \\ & x_1, \quad x_2 \geq 0 \end{aligned}$$

primal problem LP

$$\begin{aligned} \min \quad & 12\pi_1 + 12\pi_2 + 8\pi_3 \\ \text{subject to} \quad & -\pi_1 + \pi_2 + 3\pi_3 \geq 5 \\ & 3\pi_1 + 2\pi_2 - 2\pi_3 \geq 2 \\ & \pi_1, \quad \pi_2, \quad \pi_3 \geq 0 \end{aligned}$$

dual problem $D(LP)$

Example: primal and dual feasible solutions (2/5)

- A feasible primal solution is:

$$\bar{x}_1 = \frac{12}{5}, \quad \bar{x}_2 = \frac{24}{5}$$

It provides the following lower bound on the optimal value:

$$LB(\bar{\boldsymbol{x}}) = 5 \cdot \frac{12}{5} + 2 \cdot \frac{24}{5} = \frac{108}{5}$$

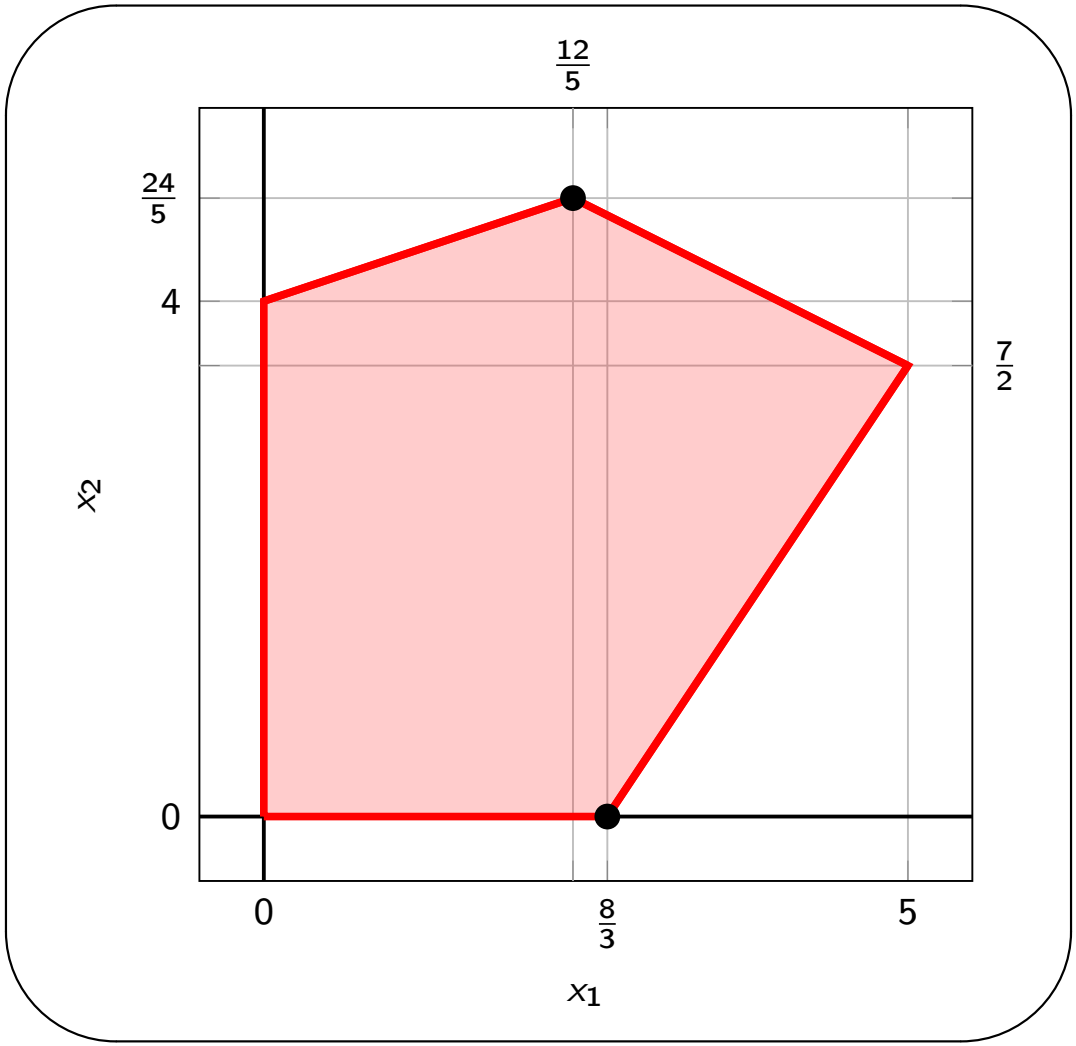
- Another feasible primal solution is:

$$\bar{x}_1 = \frac{8}{3}, \quad \bar{x}_2 = 0$$

It provides the following lower bound on the optimal value:

$$LB(\bar{\boldsymbol{x}}) = 5 \cdot \frac{8}{3} + 2 \cdot 0 = \frac{40}{3}$$

Example: primal and dual feasible solutions (3/5)



- The feasible region of the primal problem and the two feasible primal solutions

Example: primal and dual feasible solutions (4/5)

- A feasible dual solution is:

$$\bar{\pi}_1 = 0, \quad \bar{\pi}_2 = 5, \quad \bar{\pi}_3 = 0$$

It provides the following upper bound on the optimal value:

$$UB(\bar{\boldsymbol{\pi}}) = 12 \cdot 0 + 12 \cdot 5 + 8 \cdot 0 = 60$$

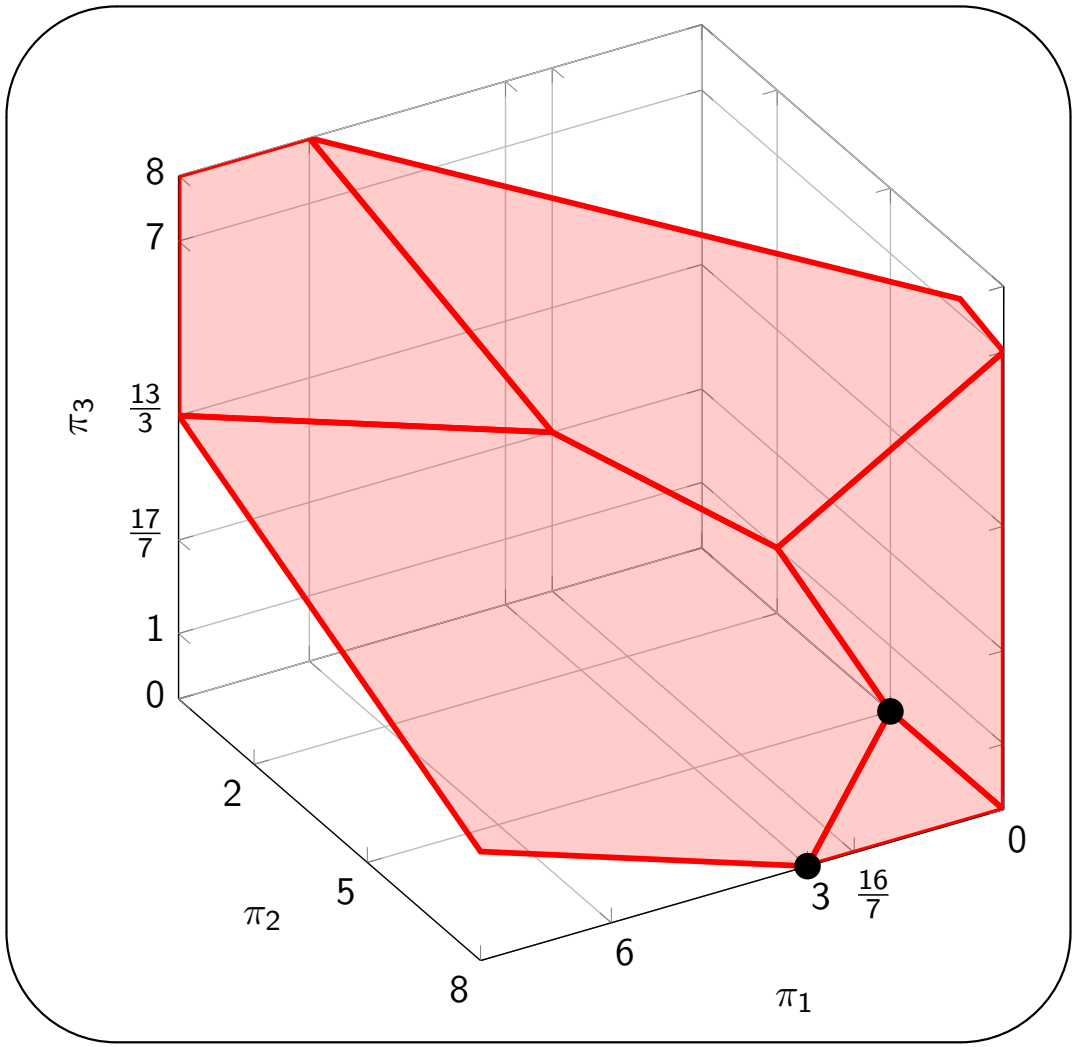
- Another feasible dual solution is:

$$\bar{\pi}_1 = 3, \quad \bar{\pi}_2 = 8, \quad \bar{\pi}_3 = 0$$

It provides the following upper bound on the optimal value:

$$UB(\bar{\boldsymbol{\pi}}) = 12 \cdot 3 + 12 \cdot 8 + 8 \cdot 0 = 132$$

Example: primal and dual feasible solutions (5/5)



- The feasible region of the dual problem and the two feasible dual solutions

Weak Duality also gives a simple way
to **prove optimality**:

a primal and a dual feasible solution
with the **same value** are both optimal.

Optimality from Weak Duality (1/3)

- Using Weak Duality, we can show that $(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n)$ is an optimal solution of the primal problem and $(\tilde{\pi}_1, \tilde{\pi}_2, \ldots, \tilde{\pi}_m)$ is an optimal solution of the dual problem if they are feasible for their respective problems and satisfy

$$\sum_{j=1}^{n} c_j \tilde{x}_j = \sum_{i=1}^{m} b_i \tilde{\pi}_i.$$

- This follows from Weak Duality, as shown in the next two slides

Optimality from Weak Duality (2/3)

1. By Weak Duality, for every feasible solution $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n)$ of the primal problem, and for the feasible dual solution $(\tilde{\pi}_1, \tilde{\pi}_2, \ldots, \tilde{\pi}_m)$, we have

$$\sum_{j=1}^{n} c_j \bar{x}_j \leq \sum_{i=1}^{m} b_i \tilde{\pi}_i.$$

If, in addition, $\sum_{j=1}^{n} c_j \tilde{x}_j = \sum_{i=1}^{m} b_i \tilde{\pi}_i$, then

$$\sum_{j=1}^{n} c_j \bar{x}_j \leq \sum_{j=1}^{n} c_j \tilde{x}_j$$

for every feasible solution $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n)$ of the primal problem. Hence, $(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n)$ is an optimal solution of the primal problem.

Optimality from Weak Duality (3/3)

2. By Weak Duality, for every feasible solution $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ of the dual problem, and for the feasible primal solution $(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n)$, we have

$$\sum_{j=1}^{n} c_j \tilde{x}_j \leq \sum_{i=1}^{m} b_i \bar{\pi}_i.$$

If, in addition, $\sum_{j=1}^{n} c_j \tilde{x}_j = \sum_{i=1}^{m} b_i \tilde{\pi}_i$, then

$$\sum_{i=1}^{m} b_i \tilde{\pi}_i \leq \sum_{i=1}^{m} b_i \bar{\pi}_i$$

for every feasible solution $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ of the dual problem. Hence, $(\tilde{\pi}_1, \tilde{\pi}_2, \ldots, \tilde{\pi}_m)$ is an optimal solution of the dual problem.

Example: primal and dual optimal solutions (1/4)

$$\begin{aligned} \max \quad & 5\,x_1 + 2\,x_2 \\ \text{subject to} \quad & -x_1 + 3\,x_2 \leq 12 \\ & x_1 + 2\,x_2 \leq 12 \\ & 3\,x_1 - 2\,x_2 \leq 8 \\ & x_1, \quad x_2 \geq 0 \end{aligned}$$

primal problem LP

$$\begin{aligned} \min \quad & 12\,\pi_1 + 12\,\pi_2 + 8\,\pi_3 \\ \text{subject to} \quad & -\pi_1 + \pi_2 + 3\,\pi_3 \geq 5 \\ & 3\,\pi_1 + 2\,\pi_2 - 2\,\pi_3 \geq 2 \\ & \pi_1, \quad \pi_2, \quad \pi_3 \geq 0 \end{aligned}$$

dual problem $D(LP)$

- We show that $\tilde{x}_1 = 5$, $\tilde{x}_2 = \frac{7}{2}$ is an optimal solution of the primal problem, and that $\tilde{\pi}_1 = 0$, $\tilde{\pi}_2 = 2$, $\tilde{\pi}_3 = 1$ is an optimal solution of the dual problem

Example: primal and dual optimal solutions (2/4)

- First, we verify that we have a feasible solution for the primal problem:

$$-\tilde{x}_1 + 3\tilde{x}_2 = -5 + 3 \cdot \frac{7}{2} = \frac{11}{2} \leq 12,$$

$$\tilde{x}_1 + 2\tilde{x}_2 = 5 + 2 \cdot \frac{7}{2} = 12 \leq 12,$$

$$3\tilde{x}_1 - 2\tilde{x}_2 = 3 \cdot 5 - 2 \cdot \frac{7}{2} = 8 \leq 8,$$

$$\tilde{x}_1 = 5 \geq 0, \qquad \tilde{x}_2 = \frac{7}{2} \geq 0.$$

Example: primal and dual optimal solutions (3/4)

- Second, we verify that we have a feasible solution for the dual problem:

$$-\tilde{\pi}_1 + \tilde{\pi}_2 + 3\tilde{\pi}_3 = -0 + 2 + 3 \cdot 1 = 5 \geq 5,$$

$$3\tilde{\pi}_1 + 2\tilde{\pi}_2 - 2\tilde{\pi}_3 = 3 \cdot 0 + 2 \cdot 2 - 2 \cdot 1 = 2 \geq 2,$$

$$\tilde{\pi}_1 = 0 \geq 0, \qquad \tilde{\pi}_2 = 2 \geq 0, \qquad \tilde{\pi}_3 = 1 \geq 0.$$

Example: primal and dual optimal solutions (4/4)

- Finally, we compute the objective function values:

$$5\tilde{x}_1 + 2\tilde{x}_2 = 5 \cdot 5 + 2 \cdot \frac{7}{2} = 25 + 7 = 32,$$

and

$$12\tilde{\pi}_1 + 12\tilde{\pi}_2 + 8\tilde{\pi}_3 = 12 \cdot 0 + 12 \cdot 2 + 8 \cdot 1 = 24 + 8 = 32.$$

- Since both solutions are feasible for their respective problems and their objective values are equal, by Weak Duality, they are **both optimal**

The following theorem, called **Strong Duality**, states that the primal problem has an optimal solution if and only if the dual problem has an optimal solution,

and in this case the optimal values of the two problems coincide.

Strong Duality

Theorem 2: Strong Duality

The primal problem has an optimal solution $(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n)$ *if and only if the dual problem has an optimal solution* $(\tilde{\pi}_1, \tilde{\pi}_2, \ldots, \tilde{\pi}_m)$*, and in this case*

$$\sum_{j=1}^{n} c_j \tilde{x}_j = \sum_{i=1}^{m} b_i \tilde{\pi}_i.$$

- The existence of an optimal solution does not imply its uniqueness: there may be several optimal solutions on one side or on both sides, but the optimal objective values are equal

Strong Duality: a necessary and sufficient condition

- Strong Duality implies that

$$\text{the primal problem has an optimal solution}$$
$$\Updownarrow$$
$$\text{the dual problem has an optimal solution}$$

- So "the primal problem has an optimal solution" is a necessary and sufficient condition for "the dual problem has an optimal solution" and vice versa

The idea of the proof

1. We first show that if the primal problem has an optimal solution, then the dual problem has an optimal solution, and the optimal values of the two problems coincide
   - This is done by using an **optimal dictionary** of the primal problem produced by the simplex algorithm to construct a feasible solution of the dual problem with the same value
   - By Weak Duality, such a dual solution is necessarily optimal, which proves one implication

2. The converse implication is obtained by applying the previous reasoning to the dual problem (equivalently, by exchanging the roles of primal and dual, since the dual of the dual is the primal problem)

Proof (1/9)

- Assume that the primal problem has an optimal solution, and consider an optimal dictionary produced by the simplex algorithm
- Its last row can be written as

$$\sum_{j=1}^{n} c_j x_j = \bar{z} + \sum_{j=1}^{n+m} \hat{c}_j x_j, \qquad \text{where} \qquad \hat{c}_j = \begin{cases} \bar{c}_j & \text{if } j \in \mathcal{N}, \\ 0 & \text{otherwise.} \end{cases}$$

- Thus, $\hat{c}_j$ is equal to the reduced cost of a variable x_j if $j \in \mathcal{N}$ (the set of indices of the non-basic variables), and it is equal to 0 otherwise

Proof (2/9)

- Rearranging the terms, we get

$$\sum_{j=1}^{n} c_j x_j = \bar{z} + \sum_{j=1}^{n} \hat{c}_j x_j + \sum_{i=1}^{m} \hat{c}_{n+i} x_{n+i},$$

  where the sum has been split into the original variables and the slack variables

- Recall that, for the slack variables, we have

$$x_{n+i} = b_i - \sum_{j=1}^{n} a_{ij} x_j, \qquad \forall i \in \{1, 2, \ldots, m\}.$$

Proof (3/9)

- Substituting these expressions into the previous identity, we obtain

$$\sum_{j=1}^{n} c_j x_j = \bar{z} + \sum_{j=1}^{n} \hat{c}_j x_j + \sum_{i=1}^{m} \hat{c}_{n+i} \left(b_i - \sum_{j=1}^{n} a_{ij} x_j \right)$$

$$= \bar{z} + \sum_{j=1}^{n} \hat{c}_j x_j + \sum_{i=1}^{m} b_i \hat{c}_{n+i} - \sum_{i=1}^{m} \hat{c}_{n+i} \sum_{j=1}^{n} a_{ij} x_j$$

Proof (4/9)

- Exchanging the two sums in the last term and grouping by variable, we obtain

$$\sum_{j=1}^{n} c_j x_j = \bar{z} + \sum_{j=1}^{n} \hat{c}_j x_j + \sum_{i=1}^{m} b_i \hat{c}_{n+i} - \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} \hat{c}_{n+i} \right) x_j \tag{A}$$

$$= \bar{z} + \sum_{i=1}^{m} b_i \hat{c}_{n+i} + \sum_{j=1}^{n} \left(\hat{c}_j - \sum_{i=1}^{m} a_{ij} \hat{c}_{n+i} \right) x_j$$

Proof (5/9)

- Since (A) is an identity in the variables $x_1, x_2, \ldots, x_n$, it holds for every choice of these variables
- Therefore, the constant terms on the two sides must coincide, and the coefficients of each variable must also coincide. Hence,

$$\bar{z} + \sum_{i=1}^{m} b_i \hat{c}_{n+i} = 0 \qquad \Longrightarrow \qquad \bar{z} = -\sum_{i=1}^{m} b_i \hat{c}_{n+i}, \qquad \text{(B)}$$

and

$$c_j = \hat{c}_j - \sum_{i=1}^{m} a_{ij} \hat{c}_{n+i}, \qquad \forall j \in \{1, 2, \ldots, n\}. \qquad \text{(C)}$$

Proof (6/9)

- Now we define

$$\tilde{\pi}_i = -\hat{c}_{n+i}, \qquad \forall i \in \{1, 2, \ldots, m\}, \tag{D}$$

and we show that $(\tilde{\pi}_1, \tilde{\pi}_2, \ldots, \tilde{\pi}_m)$ is an optimal solution of the dual problem with the same value as the primal one

- Since the dictionary is optimal, we have $\hat{c}_j \leq 0$ for all $j \in \{1, 2, \ldots, n+m\}$. Hence, from (D) we obtain

$$\tilde{\pi}_i = -\hat{c}_{n+i} \geq 0, \qquad \forall i \in \{1, 2, \ldots, m\},$$

Proof (7/9)

- From (C) we get

$$\sum_{i=1}^{m} a_{ij}\tilde{\pi}_i = c_j - \hat{c}_j \geq c_j, \qquad \forall j \in \{1, 2, \ldots, n\}.$$

  Therefore, $(\tilde{\pi}_1, \tilde{\pi}_2, \ldots, \tilde{\pi}_m)$ is **feasible** for the dual problem

- Moreover, since $(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n)$ is an optimal solution of the primal problem, its objective value is equal to the optimal value $\bar{z}$ of the primal problem. Hence, by (B), we have

$$\sum_{j=1}^{n} c_j\tilde{x}_j = \bar{z} = \sum_{i=1}^{m} b_i\tilde{\pi}_i.$$

Proof (8/9)

- Finally, Weak Duality implies that for every feasible dual solution $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$, $\sum_{j=1}^{n} c_j \tilde{x}_j \leq \sum_{i=1}^{m} b_i \bar{\pi}_i$
- Since $\sum_{j=1}^{n} c_j \tilde{x}_j = \sum_{i=1}^{m} b_i \tilde{\pi}_i$, it follows that

$$\sum_{i=1}^{m} b_i \tilde{\pi}_i \leq \sum_{i=1}^{m} b_i \bar{\pi}_i$$

for every feasible dual solution: $(\tilde{\pi}_1, \tilde{\pi}_2, \ldots, \tilde{\pi}_m)$ is an **optimal** solution of the dual problem

Proof (9/9)

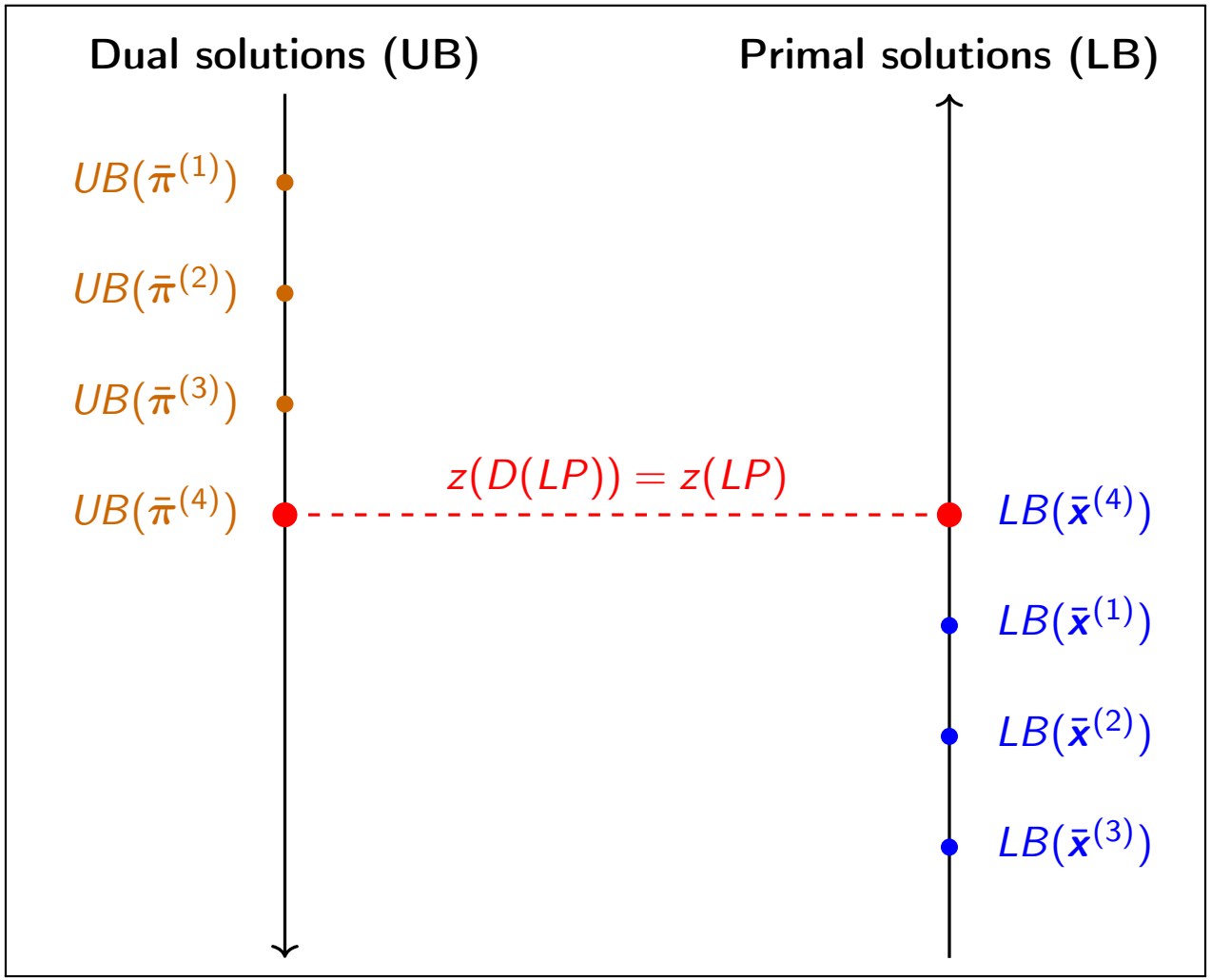
- The converse implication is obtained by applying the previous reasoning to the dual problem (after rewriting it in the corresponding canonical form, if needed): since the dual of the dual problem is the primal problem, if the dual problem has an optimal solution, then the primal problem also has an optimal solution
- Hence, the primal problem has an optimal solution if and only if the dual problem has an optimal solution, and the two optimal values coincide

□

Primal and dual bounds

- The next figure illustrates the relationship between primal and dual bounds for a **maximization** LP problem
- As an illustrative example, it considers four primal feasible solutions $\bar{\boldsymbol{x}}^{(1)}, \bar{\boldsymbol{x}}^{(2)}, \bar{\boldsymbol{x}}^{(3)}, \bar{\boldsymbol{x}}^{(4)}$ and four dual feasible solutions $\bar{\boldsymbol{\pi}}^{(1)}, \bar{\boldsymbol{\pi}}^{(2)}, \bar{\boldsymbol{\pi}}^{(3)}, \bar{\boldsymbol{\pi}}^{(4)}$
- Primal feasible solutions provide **lower bounds** on the optimal value, while dual feasible solutions provide **upper bounds** on the optimal value, by the Weak Duality theorem
- The fourth solution of each sequence attains the optimal value, and by the Strong Duality theorem the two bounds meet: $z(\, LP\,) = z(\, D(LP)\,)$

Primal and dual bounds: the figure



The proof of Strong Duality is constructive:
let us read an optimal dual solution
directly from an **optimal dictionary** of the guiding example.

Example: optimal dual solutions from dictionaries (1/5)

- Let us consider the primal problem of the guiding example (where $n = 2$ and $m = 3$). The standard form of the primal problem is

$$
\begin{array}{rrrrrrrrrrrrrrr}
\max & 5 & x_1 & + & 2 & x_2 & & & & & & & & & \\
\text{subject to} & - & x_1 & + & 3 & x_2 & + & x_3 & & & & & & = & 12 \\
 & & x_1 & + & 2 & x_2 & & & + & x_4 & & & & = & 12 \\
 & 3 & x_1 & - & 2 & x_2 & & & & & + & x_5 & = & 8 \\
 & & x_1, & & & x_2, & & x_3, & & x_4, & & x_5 & \geq & 0
\end{array}
$$

- x_3, x_4 and x_5 are the slack variables associated with the three constraints

Example: optimal dual solutions from dictionaries (2/5)

- An optimal dictionary reads as follows:

$$\begin{cases} x_1 = 5 - \frac{1}{4} x_4 - \frac{1}{4} x_5 \\ x_2 = \frac{7}{2} - \frac{3}{8} x_4 + \frac{1}{8} x_5 \\ x_3 = \frac{13}{2} + \frac{7}{8} x_4 - \frac{5}{8} x_5 \\ z = 32 - 2 x_4 - x_5 \end{cases}$$

- The basic variables are x_1, x_2 and x_3, and the non-basic variables are x_4 and x_5
- The optimal primal solution is $\tilde{x}_1 = 5$, $\tilde{x}_2 = \frac{7}{2}$, $\tilde{x}_3 = \frac{13}{2}$, $\tilde{x}_4 = 0$ and $\tilde{x}_5 = 0$, and the optimal value is 32

Example: optimal dual solutions from dictionaries (3/5)

- The dual problem is

$$\begin{array}{rrcrcrcl} \min & 12\,\pi_1 & + & 12\,\pi_2 & + & 8\,\pi_3 & & \\ \text{subject to} & -\,\pi_1 & + & \pi_2 & + & 3\,\pi_3 & \geq & 5 \\ & 3\,\pi_1 & + & 2\,\pi_2 & - & 2\,\pi_3 & \geq & 2 \\ & \pi_1, & & \pi_2, & & \pi_3 & \geq & 0 \end{array}$$

- The values $\hat{c}_j$ with $j \in \{1, 2, 3, 4, 5\}$ used in the proof of the Strong Duality theorem are:

$$\hat{c}_1 = 0, \quad \hat{c}_2 = 0, \quad \hat{c}_3 = 0, \quad \hat{c}_4 = -2, \quad \hat{c}_5 = -1$$

Example: optimal dual solutions from dictionaries (4/5)

- The optimal dual solution can be obtained by assigning the values $\hat{c}_j$ of the slack variables to the corresponding dual variables with reversed signs, as in (D):

$$\tilde{\pi}_1 = -\hat{c}_3 = 0, \quad \tilde{\pi}_2 = -\hat{c}_4 = 2, \quad \tilde{\pi}_3 = -\hat{c}_5 = 1$$

- The solution is feasible for the dual problem, since $\tilde{\pi}_1$, $\tilde{\pi}_2$ and $\tilde{\pi}_3$ are non-negative and

$$-\tilde{\pi}_1 + \tilde{\pi}_2 + 3\tilde{\pi}_3 = -0 + 2 + 3 \cdot 1 = 5 \geq 5,$$

$$3\tilde{\pi}_1 + 2\tilde{\pi}_2 - 2\tilde{\pi}_3 = 3 \cdot 0 + 2 \cdot 2 - 2 \cdot 1 = 2 \geq 2.$$

Example: optimal dual solutions from dictionaries (5/5)

- The objective function value of this solution is:

$$12 \cdot 0 + 12 \cdot 2 + 8 \cdot 1 = 32$$

- It is equal to the optimal value 32 of the primal problem: the dual solution read from the optimal dictionary is **optimal**, as stated by the Strong Duality theorem

- It is the same optimal dual solution $\tilde{\pi}_1 = 0$, $\tilde{\pi}_2 = 2$, $\tilde{\pi}_3 = 1$ that we checked with Weak Duality

Summary: the dual problem (1/2)

Dual problem:

- One dual variable $\pi_i \geq 0$ for each primal constraint, one dual constraint $\sum_{i=1}^{m} a_{ij}\pi_i \geq c_j$ for each primal variable, objective $\min \sum_{i=1}^{m} b_i\pi_i$
- It looks for the best upper bound on $z(LP)$ given by non-negative linear combinations of the constraints
- The dual of the dual is the primal problem

Weak Duality:

- Every primal feasible value is at most every dual feasible value
- Dual feasible solutions give upper bounds, primal feasible solutions give lower bounds
- A primal and a dual feasible solution with the same value are both optimal

Summary: the dual problem (2/2)

Strong Duality:

- The primal problem has an optimal solution if and only if the dual problem has one, and the two optimal values coincide
- The proof builds an optimal dual solution from an optimal dictionary of the primal problem: $\tilde{\pi}_i = -\hat{c}_{n+i}$, the reduced costs of the slack variables with reversed signs

Guiding example:

- $\tilde{x} = (5, \frac{7}{2})$ and $\tilde{\pi} = (0, 2, 1)$ are optimal, with value 32

Primal and dual problems: possible outcomes

Introduction: Primal and dual problems: possible outcomes

- The Weak Duality and the Strong Duality theorems relate the optimal values of the primal problem and of the dual problem whenever both admit an optimal solution
- We now describe what happens when one of the two problems is **infeasible** or **unbounded**
- Only four of the nine possible combinations can occur: we prove it and we see an example of each new case
- The result can be used to **certify** the outcome of an LP problem

Unbounded problems and infeasible problems

Corollary 1: unbounded problems and infeasible problems

For a pair of primal and dual problems:

1. *if the primal problem is **unbounded**, then the dual problem is **infeasible**;*
2. *if the dual problem is **unbounded**, then the primal problem is **infeasible**.*

- The idea of the proof is that one single feasible solution of one of the two problems would bound the objective value of the other one, by Weak Duality

Proof (1/2)

- We prove the first point; the second one is obtained by exchanging the roles of the two problems
- Assume, by contradiction, that the primal problem is unbounded and that the dual problem admits a feasible solution $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$

Proof (2/2)

- By Weak Duality, for every feasible solution $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n)$ of the primal problem we have

$$\sum_{j=1}^{n} c_j \bar{x}_j \leq \underbrace{\sum_{i=1}^{m} b_i \bar{\pi}_i}_{=M},$$

  where M is a fixed real number, since the dual feasible solution is fixed

- Accordingly, no feasible solution of the primal problem has an objective value larger than M, which contradicts the assumption that the primal problem is unbounded

□

Possible outcomes

Theorem 3: possible outcomes of a pair of primal and dual problems

For a pair of primal and dual problems, exactly one of the following four cases occurs:

1. *both problems admit an **optimal solution**, and the two optimal values coincide;*
2. *the primal problem is **unbounded** and the dual problem is **infeasible**;*
3. *the primal problem is **infeasible** and the dual problem is **unbounded**;*
4. *both problems are **infeasible**.*

Proof (1/3)

- By the Fundamental Theorem of Linear Programming, each of the two problems is infeasible, is unbounded, or admits an optimal solution
- Then there are **nine** possible combinations: we show that **five** of them cannot occur

primal problem	**dual problem**		
	optimal solution	unbounded	infeasible
optimal solution	?	?	?
unbounded	?	?	?
infeasible	?	?	?

Proof (2/3)

- If the primal problem admits an optimal solution, then by Strong Duality the dual problem admits an optimal solution as well
- This excludes the two combinations in which the primal problem admits an optimal solution and the dual problem is unbounded or infeasible
- Symmetrically, the two combinations in which the dual problem admits an optimal solution and the primal problem is unbounded or infeasible are excluded as well

Proof (3/3)

- If the primal problem is unbounded, then by the previous corollary the dual problem is infeasible
- This excludes the combination in which both problems are unbounded
- The four remaining combinations are the ones of the statement

□

Possible outcomes: the table

- The possible outcomes of a pair of primal and dual problems are summarized in the following table, where the four cases of the theorem are marked by ✓:

primal problem LP	**dual problem** $D(LP)$ optimal solution	unbounded	infeasible
optimal solution	✓		
unbounded			✓
infeasible		✓	✓

- In particular, an infeasible primal problem may have either an unbounded or an infeasible dual problem: the infeasibility of the primal problem alone does not determine the outcome of the dual problem

Let us see an example of each of the three cases in which one of the two problems is infeasible.

Example: unbounded primal and infeasible dual (1/3)

- Consider the following pair of primal and dual problems, with $n = 2$ and $m = 2$:

$$\begin{array}{rrcrcl} \max & x_1 & + & x_2 & & \\ \text{subject to} & -x_1 & + & x_2 & \leq & 1 \\ & x_1 & - & x_2 & \leq & 1 \\ & x_1, & & x_2 & \geq & 0 \end{array} \qquad \begin{array}{rrcrcl} \min & \pi_1 & + & \pi_2 & & \\ \text{subject to} & -\pi_1 & + & \pi_2 & \geq & 1 \\ & \pi_1 & - & \pi_2 & \geq & 1 \\ & \pi_1, & & \pi_2 & \geq & 0 \end{array}$$

- The primal problem is **unbounded**: for every value $t \geq 0$, the point $\bar{x}_1 = \bar{x}_2 = t$ is a feasible solution of objective value $2\,t$

Example: unbounded primal and infeasible dual (2/3)

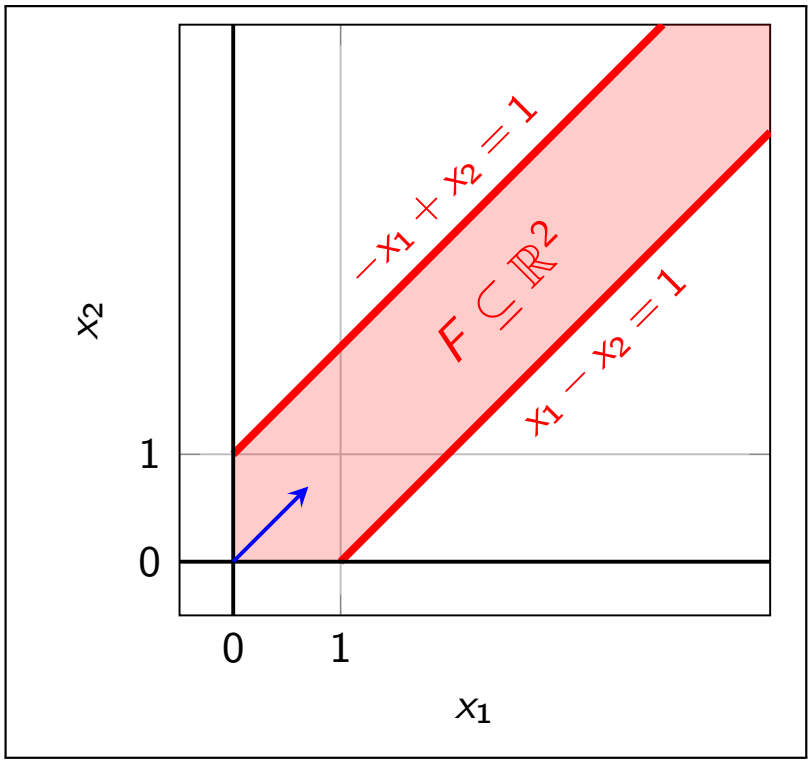
- The dual problem is **infeasible**: by summing its two constraints we obtain

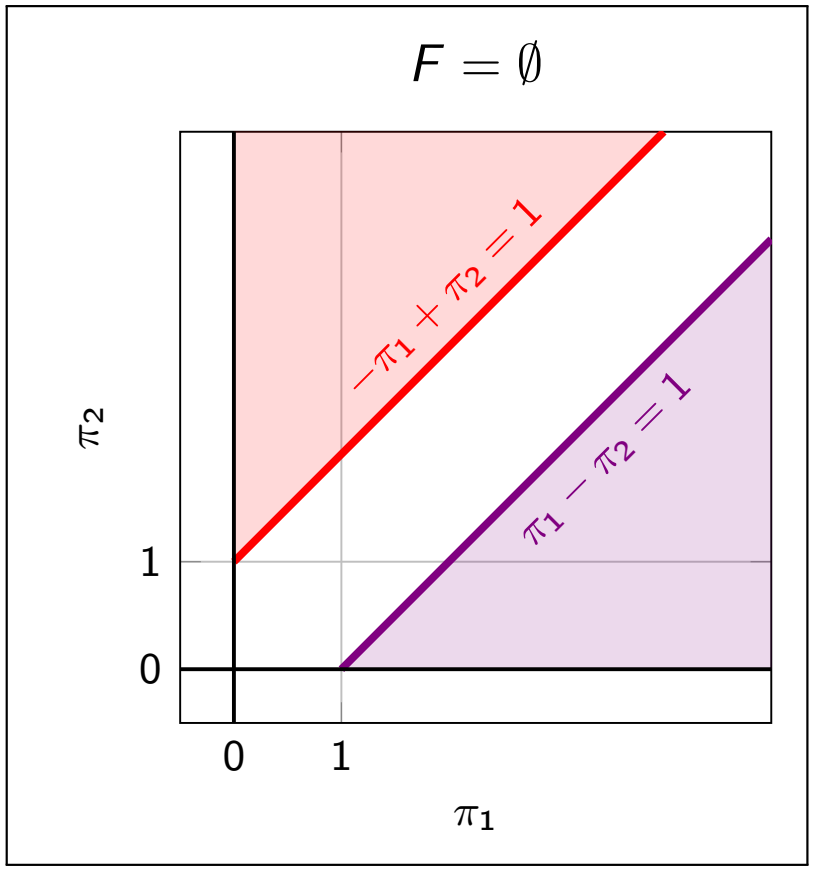
$$\underbrace{(-\pi_1 + \pi_2)}_{\geq 1} + \underbrace{(\pi_1 - \pi_2)}_{\geq 1} = 0 \geq 2,$$

  which is never satisfied

- As stated by the corollary, the primal problem is unbounded and the dual problem is infeasible

Example: unbounded primal and infeasible dual (3/3)





- On the left the feasible region of the primal problem, on the right the two half-planes defined by the dual constraints, whose intersection is empty

Example: infeasible primal and unbounded dual (1/3)

- Consider the following pair of primal and dual problems, with $n = 2$ and $m = 2$:

$$\begin{array}{rrcrcrl} \max & x_1 & + & x_2 & & & \\ \text{subject to} & x_1 & + & x_2 & \leq & 2 & \\ & -x_1 & - & x_2 & \leq & -4 & \\ & x_1, & & x_2 & \geq & 0 & \end{array} \qquad \begin{array}{rrcrcrl} \min & 2\,\pi_1 & - & 4\,\pi_2 & & & \\ \text{subject to} & \pi_1 & - & \pi_2 & \geq & 1 & \\ & \pi_1 & - & \pi_2 & \geq & 1 & \\ & \pi_1, & & \pi_2 & \geq & 0 & \end{array}$$

- The primal problem is **infeasible**: the first constraint imposes $x_1 + x_2 \leq 2$ and the second one imposes $x_1 + x_2 \geq 4$

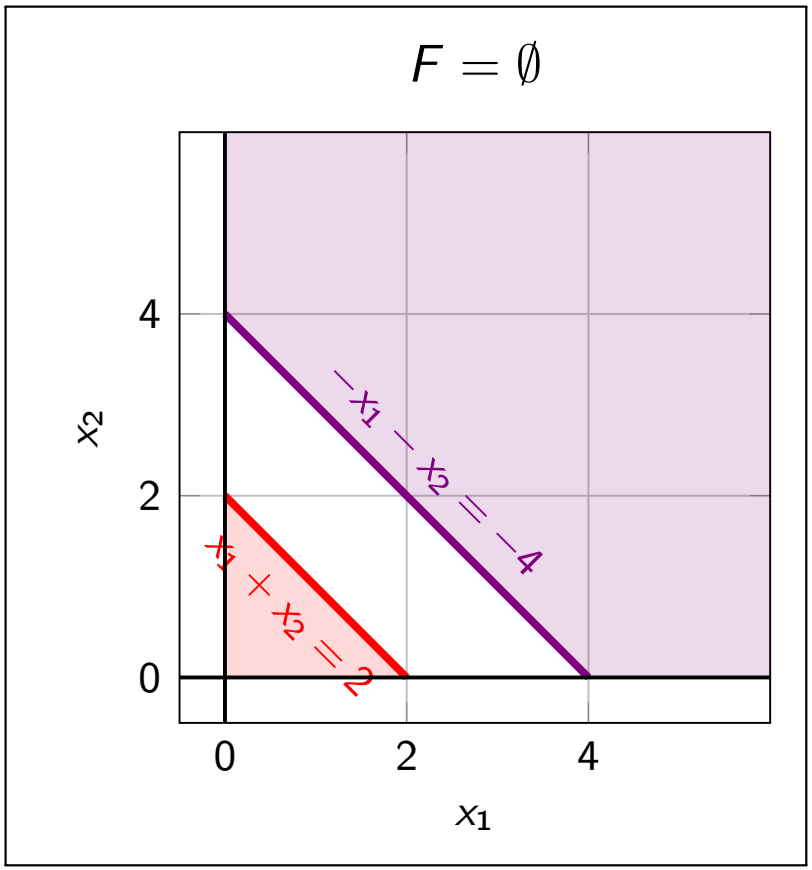
Example: infeasible primal and unbounded dual (2/3)

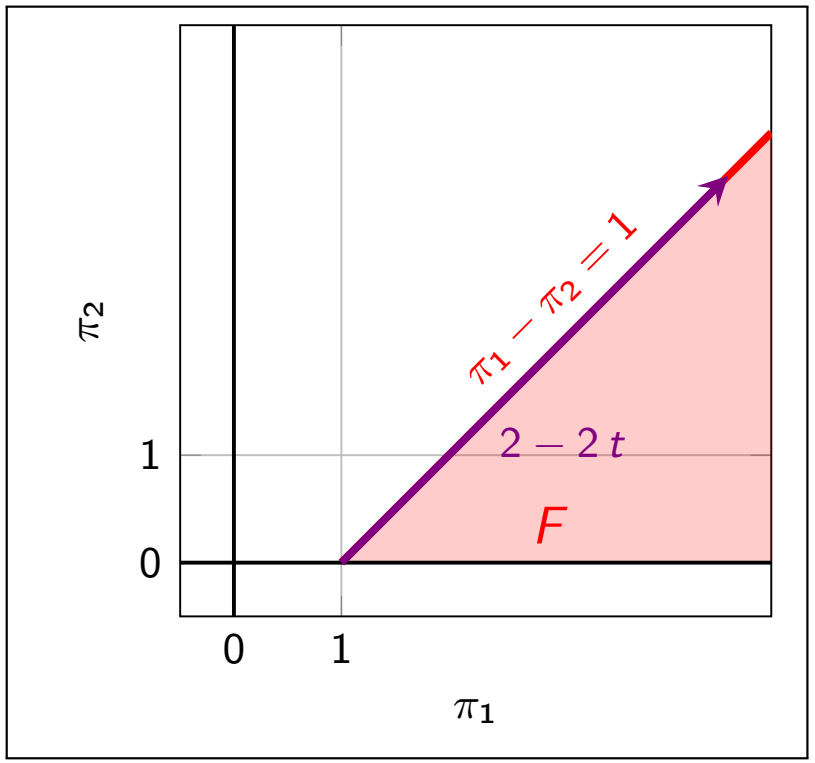
- The dual problem is **unbounded**: for every value $t \geq 0$, the point $\bar{\pi}_1 = 1 + t$ and $\bar{\pi}_2 = t$ is a feasible solution, since $\bar{\pi}_1 - \bar{\pi}_2 = 1 \geq 1$
- Its objective value is

$$2\,(1 + t) - 4\,t = 2 - 2\,t,$$

  which tends to $-\infty$ as t grows
- Since the dual problem is a minimization problem, it is unbounded

Example: infeasible primal and unbounded dual (3/3)





- On the left the two half-planes defined by the primal constraints, whose intersection is empty; on the right the feasible region of the dual problem and, in violet, the family of feasible solutions along which its objective value tends to $-\infty$

Example: infeasible primal and infeasible dual (1/3)

- Consider the following pair of primal and dual problems, with $n = 2$ and $m = 2$:

$$\begin{array}{rrcrcrl} \max & x_1 & + & x_2 & & & \\ \text{subject to} & x_1 & - & x_2 & \leq & 1 \\ & -x_1 & + & x_2 & \leq & -2 \\ & x_1, & & x_2 & \geq & 0 \end{array} \qquad \begin{array}{rrcrcrl} \min & \pi_1 & - & 2\,\pi_2 & & & \\ \text{subject to} & \pi_1 & - & \pi_2 & \geq & 1 \\ & -\pi_1 & + & \pi_2 & \geq & 1 \\ & \pi_1, & & \pi_2 & \geq & 0 \end{array}$$

- The primal problem is **infeasible**: the two constraints impose $x_1 - x_2 \leq 1$ and $x_1 - x_2 \geq 2$

Example: infeasible primal and infeasible dual (2/3)

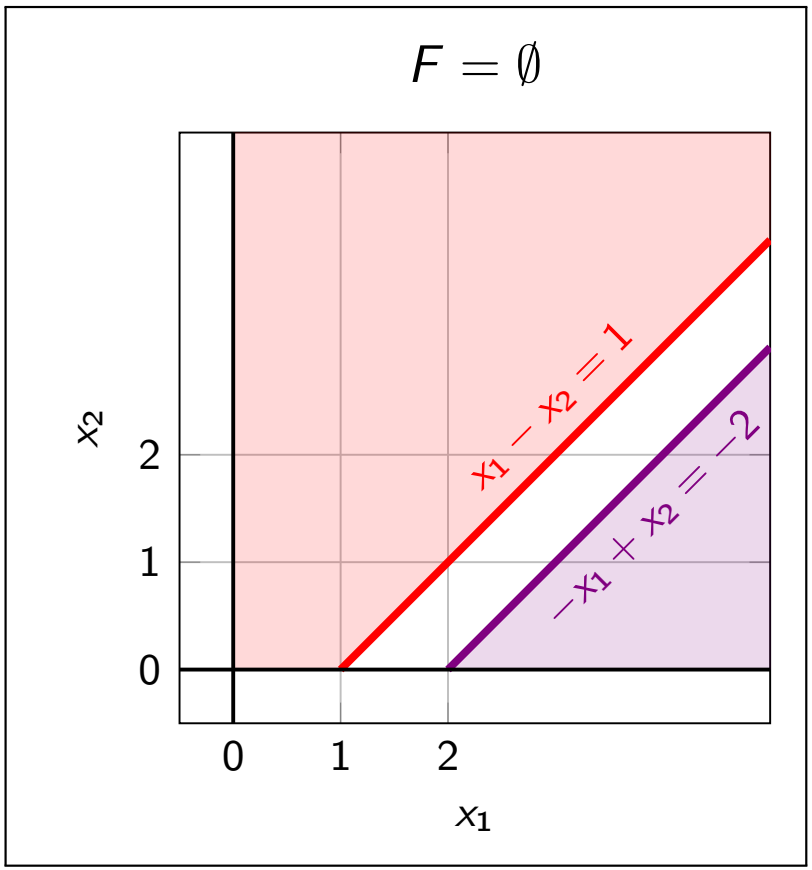
- The dual problem is **infeasible** as well: by summing its two constraints we obtain

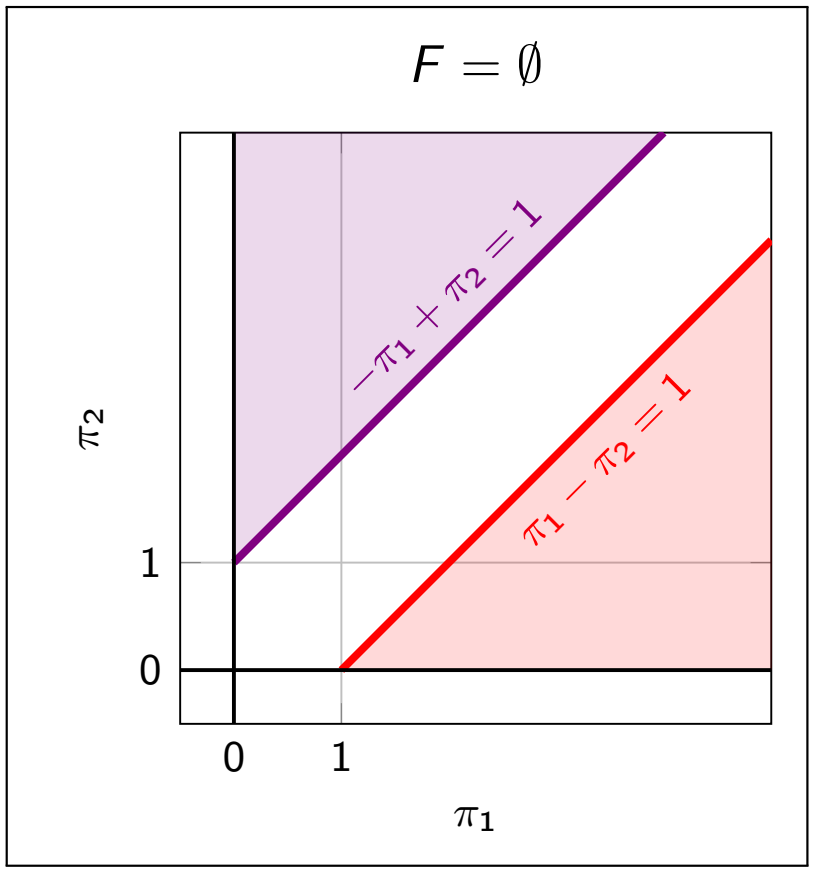
$$\underbrace{(\pi_1 - \pi_2)}_{\geq 1} + \underbrace{(-\pi_1 + \pi_2)}_{\geq 1} = 0 \geq 2,$$

  which is never satisfied

- Both problems are infeasible: the fourth case of the theorem does occur

Example: infeasible primal and infeasible dual (3/3)





- On the left the primal problem and on the right the dual problem: both feasible regions are empty

Certifying the outcome of an LP problem

- The theorem on the possible outcomes can be used to **certify** the outcome of an LP problem:

1. one feasible solution of the dual problem proves that the primal problem is not unbounded;
2. one feasible solution of the primal problem proves that the dual problem is not unbounded;
3. one feasible solution of each of the two problems proves that both admit an optimal solution.

Summary: possible outcomes

Unbounded and infeasible problems:

- If one of the two problems is unbounded, the other one is infeasible
- One feasible solution of a problem bounds the objective value of the other one

Four possible cases:

- Both optimal, with equal optimal values
- Primal unbounded and dual infeasible; primal infeasible and dual unbounded
- Both infeasible
- An infeasible primal problem alone does not determine the outcome of the dual problem

Complementary Slackness

Introduction: Complementary Slackness

- Strong Duality says that the optimal values coincide; **Complementary Slackness** says how optimal primal and dual solutions are linked, constraint by constraint
- If a constraint has a slack, the corresponding variable of the other problem is zero
- It gives an **optimality check**: a feasible primal solution is optimal if and only if we can build a suitable dual solution
- At the end, we write the primal and the dual problems in **matrix notation**

The following theorem, called **Complementary Slackness**, states that a feasible solution of the primal problem and a feasible solution of the dual problem are both optimal if and only if the Complementary Slackness Conditions hold.

Complementary Slackness

Theorem 4: Complementary Slackness

A feasible solution $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n)$ *of the primal problem and a feasible solution* $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ *of the dual problem are optimal solutions of their respective problems if and only if the following conditions hold:*

$$\sum_{j=1}^{n} a_{ij}\bar{x}_j = b_i \quad \text{or} \quad \bar{\pi}_i = 0, \qquad \forall i \in \{1, \ldots, m\}, \quad \text{(CS1)}$$

$$\sum_{i=1}^{m} a_{ij}\bar{\pi}_i = c_j \quad \text{or} \quad \bar{x}_j = 0, \qquad \forall j \in \{1, \ldots, n\}. \quad \text{(CS2)}$$

- In (CS1) and (CS2), the "or" is **inclusive**, that is, both conditions may hold simultaneously

Complementary Slackness Conditions

Definition 2: Complementary Slackness Conditions

Conditions (CS1) and (CS2) are called **Complementary Slackness Conditions** *and they can be equivalently rewritten as:*

$$\bar{\pi}_i \left(\sum_{j=1}^{n} a_{ij} \bar{x}_j - b_i \right) = 0, \qquad \forall i \in \{1, \ldots, m\},$$

$$\bar{x}_j \left(c_j - \sum_{i=1}^{m} a_{ij} \bar{\pi}_i \right) = 0, \qquad \forall j \in \{1, \ldots, n\}.$$

- A product is zero if and only if at least one of its two factors is zero

A necessary and sufficient condition

- Complementary Slackness implies that

  a pair of primal and dual feasible solutions are optimal

  $$\Updownarrow$$

  the Complementary Slackness Conditions hold

- So "the Complementary Slackness Conditions hold" is a necessary and sufficient condition for "a pair of feasible primal and dual solutions are optimal" and vice versa

Active constraints

Definition 3: active constraints

A primal or a dual constraint is called:

1. ***active*** *if it is satisfied at equality by a given solution (there is no slack);*
2. ***not active*** *if it is not satisfied at equality by a given solution (there is a slack).*

- Condition (CS1) implies that if a **primal constraint** is not active, i.e., there is a slack, the corresponding **dual variable** must be zero
- Condition (CS2) implies that if a **dual constraint** is not active, i.e., there is a slack, the corresponding **primal variable** must be zero

Example: complementary slackness (1/4)

- Consider the following pair of primal and dual LP problems:

$$\begin{aligned} \max \quad & 5\,x_1 + 2\,x_2 \\ \text{subject to} \quad & -x_1 + 3\,x_2 \leq 12 \\ & x_1 + 2\,x_2 \leq 12 \\ & 3\,x_1 - 2\,x_2 \leq 8 \\ & x_1, \quad x_2 \geq 0 \end{aligned}$$

primal problem LP

$$\begin{aligned} \min \quad & 12\,\pi_1 + 12\,\pi_2 + 8\,\pi_3 \\ \text{subject to} \quad & -\pi_1 + \pi_2 + 3\,\pi_3 \geq 5 \\ & 3\,\pi_1 + 2\,\pi_2 - 2\,\pi_3 \geq 2 \\ & \pi_1, \quad \pi_2, \quad \pi_3 \geq 0 \end{aligned}$$

dual problem $D(LP)$

- An optimal primal solution is $\tilde{x}_1 = 5$, $\tilde{x}_2 = \frac{7}{2}$

Example: complementary slackness (2/4)

- Primal slacks:

$$12 - (-\tilde{x}_1 + 3\tilde{x}_2) = 12 - \left(-5 + 3 \cdot \frac{7}{2}\right) = \frac{13}{2} > 0 \quad \Rightarrow \quad \tilde{\pi}_1 = 0,$$

$$12 - (\tilde{x}_1 + 2\tilde{x}_2) = 12 - (5 + 7) = 0 \quad \Rightarrow \quad \tilde{\pi}_2 \text{ may be } > 0,$$

$$8 - (3\tilde{x}_1 - 2\tilde{x}_2) = 8 - (15 - 7) = 0 \quad \Rightarrow \quad \tilde{\pi}_3 \text{ may be } > 0.$$

- The first primal constraint is not active, i.e., there is a slack. So the corresponding dual variable $\tilde{\pi}_1$ must be zero
- The second and third primal constraints are active, i.e., there are no slacks. So the corresponding dual variables $\tilde{\pi}_2$ and $\tilde{\pi}_3$ are free to be positive

Example: complementary slackness (3/4)

- An optimal dual solution is

$$\tilde{\pi}_1 = 0, \qquad \tilde{\pi}_2 = 2, \qquad \tilde{\pi}_3 = 1.$$

- Dual slacks:

$$(-\tilde{\pi}_1 + \tilde{\pi}_2 + 3\tilde{\pi}_3) - 5 = (-0 + 2 + 3 \cdot 1) - 5 = 0$$

$$\Rightarrow \quad \tilde{x}_1 \text{ may be } > 0,$$

$$(3\tilde{\pi}_1 + 2\tilde{\pi}_2 - 2\tilde{\pi}_3) - 2 = (3 \cdot 0 + 2 \cdot 2 - 2 \cdot 1) - 2 = 0$$

$$\Rightarrow \quad \tilde{x}_2 \text{ may be } > 0.$$

Example: complementary slackness (4/4)

- All dual constraints are active, i.e., there are no slacks
- So the corresponding primal variables $\tilde{x}_1$ and $\tilde{x}_2$ are free to be positive
- Indeed, $\tilde{x}_1 = 5 > 0$ and $\tilde{x}_2 = \frac{7}{2} > 0$, while the only non-active primal constraint, the first one, has $\tilde{\pi}_1 = 0$

Geometric meaning

- The Complementary Slackness Conditions geometrically mean that, given an optimal solution $(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n)$ of a primal problem and an optimal solution $(\tilde{\pi}_1, \tilde{\pi}_2, \ldots, \tilde{\pi}_m)$ of its dual problem, we have two pairs of **perpendicular vectors**:

$$\begin{pmatrix} \tilde{\pi}_1 \\ \tilde{\pi}_2 \\ \vdots \\ \tilde{\pi}_m \end{pmatrix}, \begin{pmatrix} \sum_{j=1}^{n} a_{1j}\tilde{x}_j - b_1 \\ \sum_{j=1}^{n} a_{2j}\tilde{x}_j - b_2 \\ \vdots \\ \sum_{j=1}^{n} a_{mj}\tilde{x}_j - b_m \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_n \end{pmatrix}, \begin{pmatrix} c_1 - \sum_{i=1}^{m} a_{i1}\tilde{\pi}_i \\ c_2 - \sum_{i=1}^{m} a_{i2}\tilde{\pi}_i \\ \vdots \\ c_n - \sum_{i=1}^{m} a_{in}\tilde{\pi}_i \end{pmatrix}$$

Example: perpendicular vectors (1/4)

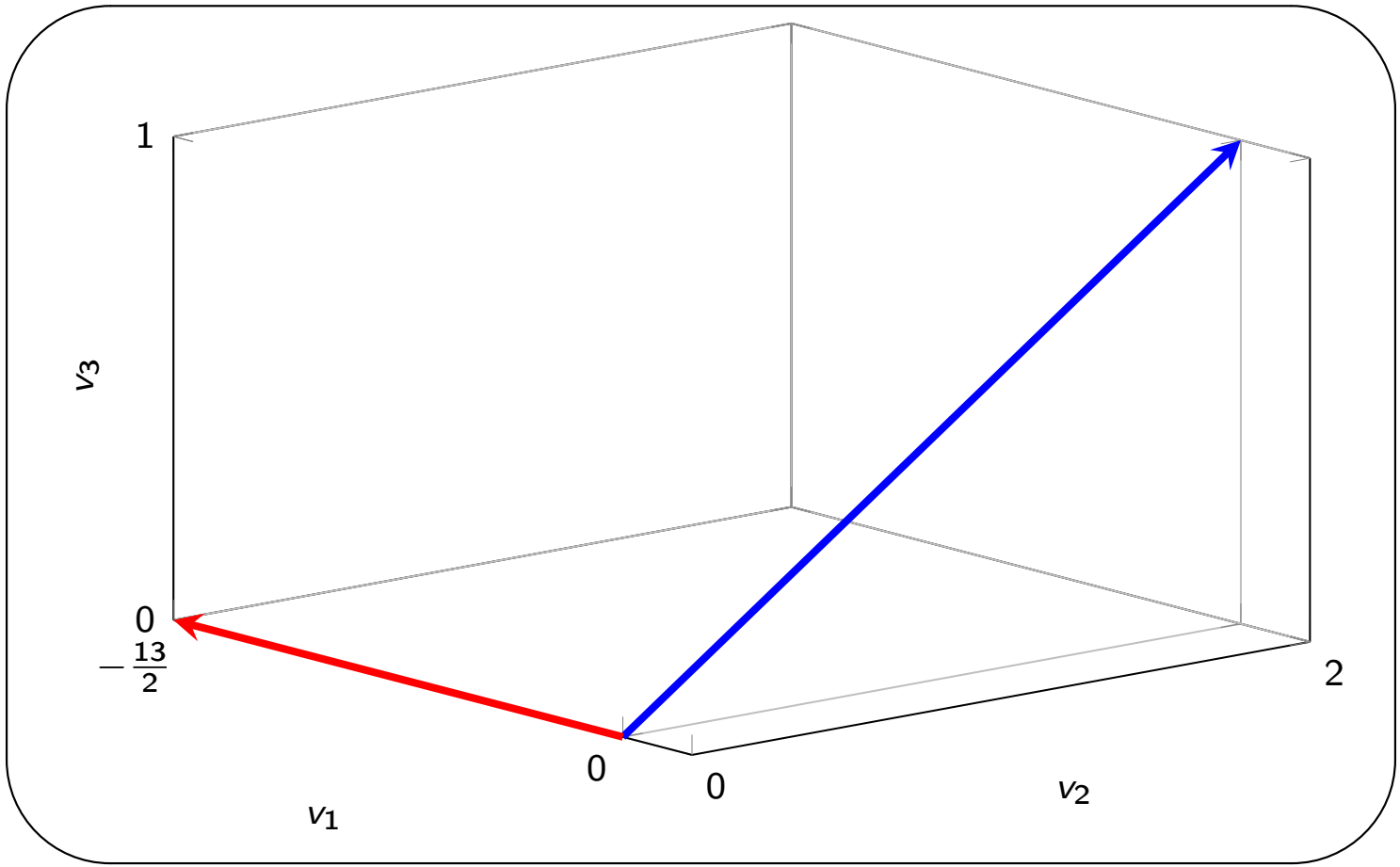
- Consider again the previous pair of primal and dual LP problems and their optimal solutions
- The first pair of perpendicular vectors (nonzero case) is

$$\begin{pmatrix} \tilde{\pi}_1 \\ \tilde{\pi}_2 \\ \tilde{\pi}_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -\tilde{x}_1 + 3\tilde{x}_2 - 12 \\ \tilde{x}_1 + 2\tilde{x}_2 - 12 \\ 3\tilde{x}_1 - 2\tilde{x}_2 - 8 \end{pmatrix} = \begin{pmatrix} -\frac{13}{2} \\ 0 \\ 0 \end{pmatrix}.$$

- Orthogonality check:

$$\begin{pmatrix} 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} -\frac{13}{2} \\ 0 \\ 0 \end{pmatrix} = 0 \cdot \left(-\frac{13}{2} \right) + 2 \cdot 0 + 1 \cdot 0 = 0.$$

Example: perpendicular vectors (2/4)



- The two vectors are drawn in the same space $\mathbb{R}^3$ with generic coordinates (v_1, v_2, v_3): in blue the optimal dual solution $\tilde{\pi}$, in red the vector of the primal constraint values minus the RHS, $\left(-\frac{13}{2}, 0, 0\right)$

Example: perpendicular vectors (3/4)

- The second pair of perpendicular vectors is

$$\begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} = \begin{pmatrix} 5 \\ \frac{7}{2} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 5 - (-\tilde{\pi}_1 + \tilde{\pi}_2 + 3\tilde{\pi}_3) \\ 2 - (3\tilde{\pi}_1 + 2\tilde{\pi}_2 - 2\tilde{\pi}_3) \end{pmatrix}$$

$$= \begin{pmatrix} 5 - (-0 + 2 + 3 \cdot 1) \\ 2 - (3 \cdot 0 + 2 \cdot 2 - 2 \cdot 1) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Example: perpendicular vectors (4/4)

- Orthogonality check:

$$\begin{pmatrix} 5 & \frac{7}{2} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0.$$

- In this case the second vector of the pair is the zero vector: all the dual constraints are active

We now prove the Complementary Slackness theorem.

The idea: look again at the two inequalities
in the proof of Weak Duality,
and ask when they hold with **equality**.

Proof (1/7)

- Since $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n)$ is a feasible solution of the primal problem and $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ is a feasible solution of the dual problem, by the same argument used in the proof of Weak Duality, we have

$$\sum_{j=1}^{n} c_j \bar{x}_j \leq \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} \bar{\pi}_i \right) \bar{x}_j$$

$$= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} \bar{x}_j \right) \bar{\pi}_i \qquad \text{(A)}$$

$$\leq \sum_{i=1}^{m} b_i \bar{\pi}_i .$$

Proof (2/7)

- The first inequality of (A) can be rewritten as

$$0 \leq \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} \bar{\pi}_i - c_j \right) \bar{x}_j,$$

and by dual feasibility and primal feasibility, for all $j \in \{1, 2, \ldots, n\}$, we have

$$\sum_{i=1}^{m} a_{ij} \bar{\pi}_i - c_j \geq 0 \qquad \text{and} \qquad \bar{x}_j \geq 0.$$

Proof (3/7)

- Each term of the sum is the product of two non-negative factors
- A sum of non-negative terms equals zero if and only if each term is zero. Therefore, equality holds in the first inequality of (A) if and only if

$$\sum_{i=1}^{m} a_{ij}\bar{\pi}_i = c_j \quad \text{or} \quad \bar{x}_j = 0, \qquad \forall j \in \{1, 2, \ldots, n\},$$

that is, if and only if condition (CS2) holds

Proof (4/7)

- Similarly, the second inequality of (A) can be rewritten as

$$0 \leq \sum_{i=1}^{m} \left(b_i - \sum_{j=1}^{n} a_{ij} \bar{x}_j \right) \bar{\pi}_i,$$

and by primal feasibility and dual feasibility, for all $i \in \{1, 2, \ldots, m\}$, we have

$$b_i - \sum_{j=1}^{n} a_{ij} \bar{x}_j \geq 0 \qquad \text{and} \qquad \bar{\pi}_i \geq 0.$$

Proof (5/7)

- Each term of the sum is the product of two non-negative factors
- A sum of non-negative terms equals zero if and only if each term is zero. Therefore, equality holds in the second inequality of (A) if and only if

$$\sum_{j=1}^{n} a_{ij}\bar{x}_j = b_i \quad \text{or} \quad \bar{\pi}_i = 0, \qquad \forall i \in \{1, 2, \ldots, m\},$$

that is, if and only if condition (CS1) holds

Proof (6/7)

- Assume first that $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n)$ and $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ are optimal solutions of their respective problems
- Then, by Strong Duality,

$$\sum_{j=1}^{n} c_j \bar{x}_j = \sum_{i=1}^{m} b_i \bar{\pi}_i.$$

- Therefore, both inequalities in (A) must hold with equality, and hence conditions (CS1) and (CS2) hold

Proof (7/7)

- Conversely, assume that conditions (CS1) and (CS2) hold
- Then both inequalities in (A) hold with equality, and therefore

$$\sum_{j=1}^{n} c_j \bar{x}_j = \sum_{i=1}^{m} b_i \bar{\pi}_i.$$

- Since $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n)$ and $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ are feasible solutions, Weak Duality implies that they are optimal solutions of their respective problems

□

The following result, derived from the previous one,
allows us to check whether a feasible solution
of a primal problem is **optimal**,

by constructing (if possible) a feasible solution
of the dual problem with the same objective function value:

it is an **optimality check**.

An optimality check

Corollary 2

A feasible solution $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n)$ *of the primal problem is optimal if and only if there exists* $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ *such that:*

$$\sum_{i=1}^{m} a_{ij}\bar{\pi}_i = c_j \quad \text{if} \quad \bar{x}_j > 0, \qquad \forall j \in \{1, \ldots, n\}, \qquad \text{(a)}$$

$$\bar{\pi}_i = 0 \quad \text{if} \quad \sum_{j=1}^{n} a_{ij}\bar{x}_j < b_i, \qquad \forall i \in \{1, \ldots, m\}, \qquad \text{(b)}$$

$$\sum_{i=1}^{m} a_{ij}\bar{\pi}_i \geq c_j, \qquad \forall j \in \{1, \ldots, n\}, \qquad \text{(c)}$$

$$\bar{\pi}_i \geq 0, \qquad \forall i \in \{1, \ldots, m\}. \qquad \text{(d)}$$

Proof (1/2)

- Assume first that $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n)$ is an optimal solution of the primal problem
- By Strong Duality, there exists an optimal solution $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ of the dual problem
- Since $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ is feasible for the dual problem, it satisfies (c) and (d)
- Moreover, since the primal and dual solutions are both optimal, by Complementary Slackness they satisfy (a) and (b)
- Hence, there exists $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ satisfying (a)–(d)

Proof (2/2)

- Conversely, assume that there exists $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ satisfying (a)–(d)
- Since $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n)$ is feasible for the primal problem by hypothesis, and (c)–(d) imply that $(\bar{\pi}_1, \bar{\pi}_2, \ldots, \bar{\pi}_m)$ is feasible for the dual problem, both vectors are feasible for their respective problems
- Furthermore, (a) implies condition (CS2). Moreover, primal feasibility together with (b) implies condition (CS1)
- Therefore, the primal and dual feasible solutions satisfy the Complementary Slackness Conditions, and by the Complementary Slackness theorem they are optimal for their respective problems

Let us apply the optimality check
to two feasible solutions of the guiding example:

$$\left(5, \frac{7}{2}\right) \quad \text{and} \quad \left(\frac{12}{5}, \frac{24}{5}\right).$$

Example: optimality check (1/4)

- We check if the feasible solution $\bar{x}_1 = 5$, $\bar{x}_2 = \frac{7}{2}$ of the LP problem of the guiding example is optimal

1. Since

$$\sum_{i=1}^{m} a_{ij}\bar{\pi}_i = c_j \quad \text{if} \quad \bar{x}_j > 0, \ \forall j \in \{1, 2, \ldots, n\}$$

we build the following system of two linear equations with three variables:

$$\begin{array}{rcrcrcrcll} - & \bar{\pi}_1 & + & & \bar{\pi}_2 & + & 3\ \bar{\pi}_3 & = & 5 & (\bar{x}_1 = 5 > 0) \\ 3\ & \bar{\pi}_1 & + & 2\ & \bar{\pi}_2 & - & 2\ \bar{\pi}_3 & = & 2 & \left(\bar{x}_2 = \frac{7}{2} > 0\right) \end{array}$$

Example: optimality check (2/4)

2. Since

$$\bar{\pi}_i = 0 \qquad \text{if} \qquad \sum_{j=1}^{n} a_{ij}\bar{x}_j < b_i, \ \forall i \in \{1, 2, \ldots, m\}$$

we compute:

$$\begin{array}{rcrcll} -1 \cdot 5 & + & 3 \cdot \frac{7}{2} & = & \frac{11}{2} < 12 & \text{so} \quad \bar{\pi}_1 = 0 \\ 1 \cdot 5 & + & 2 \cdot \frac{7}{2} & = & 12 \not< 12 & \\ 3 \cdot 5 & - & 2 \cdot \frac{7}{2} & = & 8 \not< 8 & \end{array}$$

- We then solve the previous system of two equations with two variables and we get:

$$\bar{\pi}_1 = 0, \quad \bar{\pi}_2 = 2, \quad \bar{\pi}_3 = 1$$

Example: optimality check (3/4)

3. We need to check

$$\sum_{i=1}^{m} a_{ij}\bar{\pi}_i \geq c_j, \quad \forall j \in \{1, 2, \ldots, n\}$$

so we compute:

$$\begin{array}{rcrcrcl} -1 \cdot 0 & + & 1 \cdot 2 & + & 3 \cdot 1 & = & 5 \geq 5 \\ 3 \cdot 0 & + & 2 \cdot 2 & - & 2 \cdot 1 & = & 2 \geq 2 \end{array}$$

Example: optimality check (4/4)

4. We need to check

$$\bar{\pi}_i \geq 0, \quad \forall i \in \{1, 2, \ldots, m\}$$

so since

$$\bar{\pi}_1 = 0 \geq 0, \quad \bar{\pi}_2 = 2 \geq 0, \quad \bar{\pi}_3 = 1 \geq 0$$

the solution is **optimal**

Example: a second optimality check (1/4)

- We check if the feasible solution $\bar{x}_1 = \frac{12}{5}$, $\bar{x}_2 = \frac{24}{5}$ of the LP problem of the guiding example is optimal

1. Since

$$\sum_{i=1}^{m} a_{ij}\bar{\pi}_i = c_j \quad \text{if} \quad \bar{x}_j > 0, \ \forall j \in \{1, 2, \ldots, n\}$$

we build the following system of two linear equations with three variables:

$$\begin{array}{rcrcrcrcll} -\bar{\pi}_1 & + & & \bar{\pi}_2 & + & 3\bar{\pi}_3 & = & 5 & \left(\bar{x}_1 = \frac{12}{5} > 0\right) \\ 3\bar{\pi}_1 & + & 2\bar{\pi}_2 & - & 2\bar{\pi}_3 & = & 2 & & \left(\bar{x}_2 = \frac{24}{5} > 0\right) \end{array}$$

Example: a second optimality check (2/4)

2. Since

$$\bar{\pi}_i = 0 \qquad \text{if} \qquad \sum_{j=1}^{n} a_{ij}\bar{x}_j < b_i, \ \forall i \in \{1, 2, \ldots, m\}$$

we compute:

$$\begin{aligned}
-1 \cdot \tfrac{12}{5} \quad &+ \quad 3 \cdot \tfrac{24}{5} \quad = \quad 12 \not< 12 \\
1 \cdot \tfrac{12}{5} \quad &+ \quad 2 \cdot \tfrac{24}{5} \quad = \quad 12 \not< 12 \\
3 \cdot \tfrac{12}{5} \quad &- \quad 2 \cdot \tfrac{24}{5} \quad = \quad -\tfrac{12}{5} < 8 \qquad \text{so} \qquad \bar{\pi}_3 = 0
\end{aligned}$$

- We then solve the previous system of two equations with two variables and we get:

$$\bar{\pi}_1 = -\frac{8}{5}, \quad \bar{\pi}_2 = \frac{17}{5}, \quad \bar{\pi}_3 = 0$$

Example: a second optimality check (3/4)

3. We need to check

$$\sum_{i=1}^{m} a_{ij}\bar{\pi}_i \geq c_j, \quad \forall j \in \{1, 2, \ldots, n\}$$

so we compute:

$$\begin{array}{rcrcrcl} -1 \cdot \left(-\frac{8}{5}\right) & + & 1 \cdot \frac{17}{5} & + & 3 \cdot 0 & = & 5 \geq 5 \\ 3 \cdot \left(-\frac{8}{5}\right) & + & 2 \cdot \frac{17}{5} & - & 2 \cdot 0 & = & 2 \geq 2 \end{array}$$

Example: a second optimality check (4/4)

4. We need to check

$$\bar{\pi}_i \geq 0, \quad \forall i \in \{1, 2, \ldots, m\}$$

but since

$$\bar{\pi}_1 = -\frac{8}{5} < 0$$

the solution is **not optimal**

- Indeed, its value $\frac{108}{5}$ is smaller than the optimal value 32

To conclude the section, we write the primal problem and its dual problem in **matrix notation**.

Input data in matrix notation (1/2)

- The input data of an LP problem in canonical maximization form with n variables and m less-than-or-equal-to inequality constraints can be represented in matrix notation by:
  - an **objective-function coefficient vector** $\boldsymbol{c} \in \mathbb{R}^n$ of n rows
  - an **RHS vector** $\boldsymbol{b} \in \mathbb{R}^m$ of m rows
  - a **constraint matrix** $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ with m rows and n columns

Input data in matrix notation (2/2)

$$\boldsymbol{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ \vdots \\ c_n \end{pmatrix} \qquad \boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & \dots & a_{2n} \\ \vdots & \vdots & \dots & \dots & \vdots \\ a_{m1} & a_{m2} & \dots & \dots & a_{mn} \end{pmatrix} \qquad \boldsymbol{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

Variables in matrix notation

- The n primal variables can be represented by a column vector $\boldsymbol{x} \in \mathbb{R}^n$ of n rows containing the n variables
- The m dual variables can be represented by a column vector $\boldsymbol{\pi} \in \mathbb{R}^m$ of m rows containing the m variables

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{pmatrix} \qquad \boldsymbol{\pi} = \begin{pmatrix} \pi_1 \\ \pi_2 \\ \vdots \\ \pi_m \end{pmatrix}$$

Primal and dual problems in matrix notation

- Using matrix notation, a primal problem in canonical maximization form and its dual problem read as follows:

max $\boldsymbol{c}'\boldsymbol{x}$	min $\boldsymbol{b}'\boldsymbol{\pi}$
subject to $\boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b}$,	subject to $\boldsymbol{A}'\boldsymbol{\pi} \geq \boldsymbol{c}$,
$\boldsymbol{x} \geq 0$.	$\boldsymbol{\pi} \geq 0$.
primal problem LP	**dual problem** $D(LP)$

- The constraint matrix of the dual problem is the **transpose** $\boldsymbol{A}'$ of the constraint matrix of the primal problem

Example: matrix notation

- For the LP problem of the guiding example, with $n = 2$ variables and $m = 3$ constraints, the input data are:

$$\boldsymbol{c} = \begin{pmatrix} 5 \\ 2 \end{pmatrix} \quad \boldsymbol{A} = \begin{pmatrix} -1 & 3 \\ 1 & 2 \\ 3 & -2 \end{pmatrix} \quad \boldsymbol{A}' = \begin{pmatrix} -1 & 1 & 3 \\ 3 & 2 & -2 \end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix}$$

- and the variables are:

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad \boldsymbol{\pi} = \begin{pmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \end{pmatrix}$$

Summary: Complementary Slackness (1/2)

Complementary Slackness:

- A primal and a dual feasible solution are both optimal if and only if (CS1) and (CS2) hold
- If a primal constraint is not active, the corresponding dual variable is zero
- If a dual constraint is not active, the corresponding primal variable is zero
- Geometrically: two pairs of perpendicular vectors

Optimality check:

- A feasible primal solution is optimal if and only if there is a dual solution satisfying (a)–(d)
- Build the equations from the positive variables, set to zero the dual variables of the non-active constraints, solve, and check dual feasibility

Summary: Complementary Slackness (2/2)

Guiding example:

- $\left(5, \frac{7}{2}\right)$ passes the check, with $\bar{\pi} = (0, 2, 1)$
- $\left(\frac{12}{5}, \frac{24}{5}\right)$ does not pass it, since $\bar{\pi}_1 = -\frac{8}{5} < 0$

Matrix notation:

- Primal: $\max \boldsymbol{c}'\boldsymbol{x}$ subject to $\boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b}$, $\boldsymbol{x} \geq 0$
- Dual: $\min \boldsymbol{b}'\boldsymbol{\pi}$ subject to $\boldsymbol{A}'\boldsymbol{\pi} \geq \boldsymbol{c}$, $\boldsymbol{\pi} \geq 0$

Dual problem of a generic LP problem

Introduction: Dual problem of a generic LP problem

- So far the primal problem was in canonical maximization form: only $\leq$ constraints and non-negative variables
- A generic LP problem may have $\leq$, $=$ and $\geq$ constraints, and non-negative, free and nonpositive variables
- We define its dual problem directly, without first rewriting the problem in canonical form
- All the duality theorems of this chapter hold for the generic case as well

Definition 4: dual problem of a generic LP problem

Given the following generic LP problem:

$$\max \quad \sum_{j \in N} c_j \, x_j$$

$$\begin{aligned}
\textit{subject to} \quad & \sum_{j \in N} a_{ij} \, x_j \leq b_i, \quad && \forall i \in M_{\leq}, \\
& \sum_{j \in N} a_{ij} \, x_j = b_i, \quad && \forall i \in M_{=}, \\
& \sum_{j \in N} a_{ij} \, x_j \geq b_i, \quad && \forall i \in M_{\geq}, \\
& x_j \geq 0, \quad && \forall j \in N_{\geq 0}, \\
& x_j \gtreqless 0, \quad && \forall j \in N_{\gtreqless 0}, \\
& x_j \leq 0, \quad && \forall j \in N_{\leq 0},
\end{aligned}$$

...

Definition: dual problem of a generic LP problem

. . . where

1. $N_{\geq 0}$, $N_{\gtrless 0}$ *and* $N_{\leq 0}$ *are the disjoint subsets of indices of the non-negative, free and nonpositive variables, respectively, and* $N = N_{\geq 0} \cup N_{\gtrless 0} \cup N_{\leq 0}$*;*

2. $M_{\leq}$, $M_{=}$ *and* $M_{\geq}$ *are the disjoint subsets of indices of the less-than-or-equal-to, equality and greater-than-or-equal-to constraints, respectively, and* $M = M_{\leq} \cup M_{=} \cup M_{\geq}$*,*

. . .

Definition: dual problem of a generic LP problem

. . . the ***dual problem*** $D(LP)$ *of the generic LP problem is the following LP problem:*

$$
\begin{aligned}
\min \quad & \sum_{i \in M} b_i \, \pi_i \\
\text{subject to} \quad & \sum_{i \in M} a_{ij} \, \pi_i \geq c_j, \quad \forall j \in N_{\geq 0}, \\
& \sum_{i \in M} a_{ij} \, \pi_i = c_j, \quad \forall j \in N_{\gtreqless 0}, \\
& \sum_{i \in M} a_{ij} \, \pi_i \leq c_j, \quad \forall j \in N_{\leq 0}, \\
& \pi_i \geq 0, \quad \forall i \in M_{\leq}, \\
& \pi_i \gtreqless 0, \quad \forall i \in M_{=}, \\
& \pi_i \leq 0, \quad \forall i \in M_{\geq}.
\end{aligned}
$$

Reading the definition

- As in the canonical case, each primal **constraint** $i \in M$ gives one dual variable π_i, and each primal **variable** x_j, $j \in N$, gives one dual constraint
- The **type of a primal constraint** fixes the sign of its dual variable: $\leq$ gives $\pi_i \geq 0$, $=$ gives π_i free, $\geq$ gives $\pi_i \leq 0$
- The **sign of a primal variable** fixes the type of its dual constraint: $x_j \geq 0$ gives $\geq$, x_j free gives $=$, $x_j \leq 0$ gives $\leq$
- With $M = M_{\leq}$ and $N = N_{\geq 0}$ we find again the canonical case

Example: dual of a generic LP problem (1/3)

- Consider the following generic LP problem:

$$
\begin{array}{rrcrcrcrcrcrcr}
\max & 4 & x_1 & + & & x_2 & + & 5 & x_3 & + & 3 & x_4 & & \\
\text{subject to} & & x_1 & - & & x_2 & - & & x_3 & + & 3 & x_4 & \leq & 1 \\
& 5 & x_1 & + & & x_2 & + & 3 & x_3 & + & 8 & x_4 & = & 55 \\
& - & x_1 & + & 2 & x_2 & + & 3 & x_3 & - & 5 & x_4 & \geq & 3 \\
& & x_1 & + & 3 & x_2 & - & 2 & x_3 & + & 2 & x_4 & \geq & 6 \\
& & x_1, & & & x_2 & & & & & & & \geq & 0 \\
& & & & & & & & x_3 & & & & \gtrless & 0 \\
& & & & & & & & & & & x_4 & \leq & 0
\end{array}
$$

Example: dual of a generic LP problem (2/3)

- As far as the index of the variables is concerned, we have:

$$N_{\geq 0} = \{1, 2\}, \quad N_{\substack{\geq \\ \leq} 0} = \{3\}, \quad N_{\leq 0} = \{4\}, \quad N = \{1, 2, 3, 4\}$$

- As far as the index of the constraints is concerned, we have:

$$M_{\leq} = \{1\}, \quad M_{=} = \{2\}, \quad M_{\geq} = \{3, 4\}, \quad M = \{1, 2, 3, 4\}$$

- So the dual problem has 4 variables $\pi_1, \pi_2, \pi_3, \pi_4$ and 4 constraints, one for each primal variable

Example: dual of a generic LP problem (3/3)

- The dual problem $D(LP)$ of the LP problem is:

$$
\begin{array}{rrcrcrcrcrcl}
\min & & \pi_1 & + & 55\,\pi_2 & + & 3\,\pi_3 & + & 6\,\pi_4 & & \\
\text{subject to} & & \pi_1 & + & 5\,\pi_2 & - & \pi_3 & + & \pi_4 & \geq & 4 \\
& - & \pi_1 & + & \pi_2 & + & 2\,\pi_3 & + & 3\,\pi_4 & \geq & 1 \\
& - & \pi_1 & + & 3\,\pi_2 & + & 3\,\pi_3 & - & 2\,\pi_4 & = & 5 \\
& 3 & \pi_1 & + & 8\,\pi_2 & - & 5\,\pi_3 & + & 2\,\pi_4 & \leq & 3 \\
& & \pi_1 & & & & & & & \geq & 0 \\
& & & & \pi_2 & & & & & \gtreqless & 0 \\
& & & & & & \pi_3 & & & \leq & 0 \\
& & & & & & & & \pi_4 & \leq & 0
\end{array}
$$

Duality theorems for the generic case

Since an equivalent LP problem
in canonical maximization form
can be obtained from an LP problem
in generic maximization form,

Weak Duality, **Strong Duality**,
and the **Complementary Slackness** theorem
hold for the generic case as well.

Summary: dual problem of a generic LP problem

Generic LP problem:

- Constraints of type $\leq$, $=$, $\geq$ (index sets $M_{\leq}$, $M_{=}$, $M_{\geq}$) and variables non-negative, free, nonpositive (index sets $N_{\geq 0}$, $N_{\gtrless 0}$, $N_{\leq 0}$)

Its dual problem:

- Primal constraint $\leq$, $=$, $\geq$ $\rightarrow$ dual variable ≥ 0, free, ≤ 0
- Primal variable ≥ 0, free, ≤ 0 $\rightarrow$ dual constraint $\geq$, $=$, $\leq$
- Weak Duality, Strong Duality and Complementary Slackness hold for the generic case as well

Summary (1/2)

- The **dual problem** of an LP problem in canonical maximization form looks for the best upper bound on its optimal value given by non-negative linear combinations of the constraints; the dual of the dual is the primal problem
- **Weak Duality**: every primal feasible value is at most every dual feasible value; a primal and a dual feasible solution with the same value are both optimal
- **Strong Duality**: the primal problem has an optimal solution if and only if the dual problem has one, with the same optimal value; an optimal dual solution is read from an optimal dictionary of the primal problem

Summary (2/2)

- **Possible outcomes**: both optimal, primal unbounded and dual infeasible, primal infeasible and dual unbounded, or both infeasible
- **Complementary Slackness**: a primal and a dual feasible solution are both optimal if and only if every non-active constraint has a zero variable on the other side
- It gives an **optimality check** for a feasible primal solution, by building a dual solution and checking its feasibility
- In **matrix notation**: $\max \boldsymbol{c}'\boldsymbol{x}$, $\boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b}$, $\boldsymbol{x} \geq 0$ has the dual $\min \boldsymbol{b}'\boldsymbol{\pi}$, $\boldsymbol{A}'\boldsymbol{\pi} \geq \boldsymbol{c}$, $\boldsymbol{\pi} \geq 0$
- The dual problem of a **generic LP problem** follows the rules on constraint types and variable signs, and all the duality theorems hold for it as well